because it’s happening asynchronously. The program will carry on doing other things until the response comes back, at which point the response is added to the event queue and processed at the next available time.

What does an event loop in poor shape look like? Well, if a program spends a lot of time processing something synchronously, such as sorting a list of a million or more competitors, the event loop will be blocked for a long time and the program will experience a blackout while it waits for the synchronous operation to complete before doing anything else. That’s exactly what happens if you try to read a massive file synchronously,

```
dataForMyMassiveFile = fs.readFileSync 'myMassiveFile.mpg'
```

or when you try to synchronously sort a million competitors by their last name. In figure 9.2 you see an event loop in a poor state with a synchronous, blocking operation on the event loop. Notice how nothing else happens during the synchronous block and how two events that have had to wait for it to complete are handled shortly after.

Figure 9.2. Blocking the event loop stops the world.

